



Unethical military testing

These are some examples of testing gone horribly wrong. Can they ever be justified? Warning: Content may be disturbing. <http://dgit.in/ZMFZsV>



Mind reading

A researcher finds out a way to know the choice you will make before you make it—but not for evil. <http://dgit.in/ZqkFcP>

From the labs

abstract before you took the drug. Now every so often a new wonder drug comes out, which promises to make people's lives better, but in turn ends up falling flat on its face. Cocaine was thought to be the new drug to treat patients for everything ranging from morphine addiction to fatigue, and was held in high esteem and prescribed by Freud, resulting in the death of two of his patients. Amphetamines, psychostimulant addictive mood altering drugs, were introduced as a cure for daytime sleepiness and even Attention Deficit Hyperactivity Disorder(ADHD), but are now slowly being discontinued because of its dangerous rate of addiction (and the observation that its derivative methamphetamine has earned the street name 'Speed' for its murderously high rate of addiction and the associated high).

Why is Modafinil any different?

Gamers will know that feeling you get when you have been struggling with that final boss-fight for hours and hours and you finally thrash that ugly brute, that ecstasy of completing that challenge- we owe most of that feeling to this thing called dopamine. One of dopamine's effects is to excite the pleasure centres of the brain, resulting in a rewarding feeling, and this can happen with a lot of things – food, sex, Modern Warfare, and is one of the reasons we keep going back to these activities. What cocaine does is that it increases the

amount of dopamine released, and what amphetamines do is that they slow down or block the reabsorption of dopamine after it is released. So in effect, both result in the user feeling euphoric and it extends that euphoria. When the feeling dies, the user itches to be 'euphoric' again and so uses again, and now, the person is addicted.

Modafinil, supposedly, is not like that. It does not affect reabsorption of dopamine, and it results in a very weak release of dopamine. Instead, it depends more on adrenaline action to keep you alert, which is what sets it apart. Also, it does not cause peripheral nervous system stimulation (which is a good thing because amphetamines and the like do, and one of the nasty results is that our urinary passage gets blocked). This is the biggest point for supporters of Modafinil and is the reason that it is being taken as seriously.

Military Presence

Testing of Modafinil has been done on the US Air Force. A study by the Air Force Research Laboratory says that 'Modafinil decreased self-ratings of depression and anger, while improving ratings of vigour, alertness, and confidence. Benefits were most noticeable after 24 to 32 h of continuous wakefulness'. It further concludes that while the question of safety of increasing doses has not been answered yet, Modafinil should be considered to be given to the military to counter short-term fatigue. Now the US military has always been on the wrong side of the medical testing scene- going as far as giving their soldiers LSD without informing them to devastating effects. Amphetamines (some of the side effects of which are auditory hallucinations, paranoia, psychotic behaviour and confusion) were prescribed by flight surgeons to pilots, and in one documented incident these pilots in turn fired a laser-guided-bomb on a friendly Canadian training force killing four Canadians. So clearly, military usage is not an indicator of how safe Modafinil is.

So who is using it?

Apparently, everybody. Students, writers, wall-street traders, coders, entrepreneurs. A quick search on the internet gives myriad testimonials of people who are



Sci-fi thriller Limitless

regular users of it. "I was an ordinary student, but the drug let me feel intermittently extraordinary", "(Modafinil) helped me break through a writer's block I was having to complete a draft proposal for the Dalai Lama Foundation."

While some users rant on endlessly about how it has changed their life, there are some people who have different views. Erowid.org, an online community where members give accounts of their experiences with drugs, has a large number of testimonials of people who have taken Modafinil, and it has some very differing ideas. One user claimed that it gave him what he calls 'tunnel vision' because of which he wasn't able to handle multiple tasks. Another claims that she hadn't felt any noticeable improvement, only a mild headache a few hours after taking it. A 35 year old recovering alcoholic also mentions that he found himself dangerously addicted to the drug, in spite of claims that Modafinil is not addictive, and points out that an addictive personality could easily fall prey to dependence on the drug. Sebastian Marshall, a self proclaimed enthusiastic amateur, noted down the effects of taking the drug in his blog, Sebastianmarshall.com, and till now it seems to be the most un-sensationalist account of someone's experience with Modafinil. Have a look.



Enhance me oh ginkgo biloba